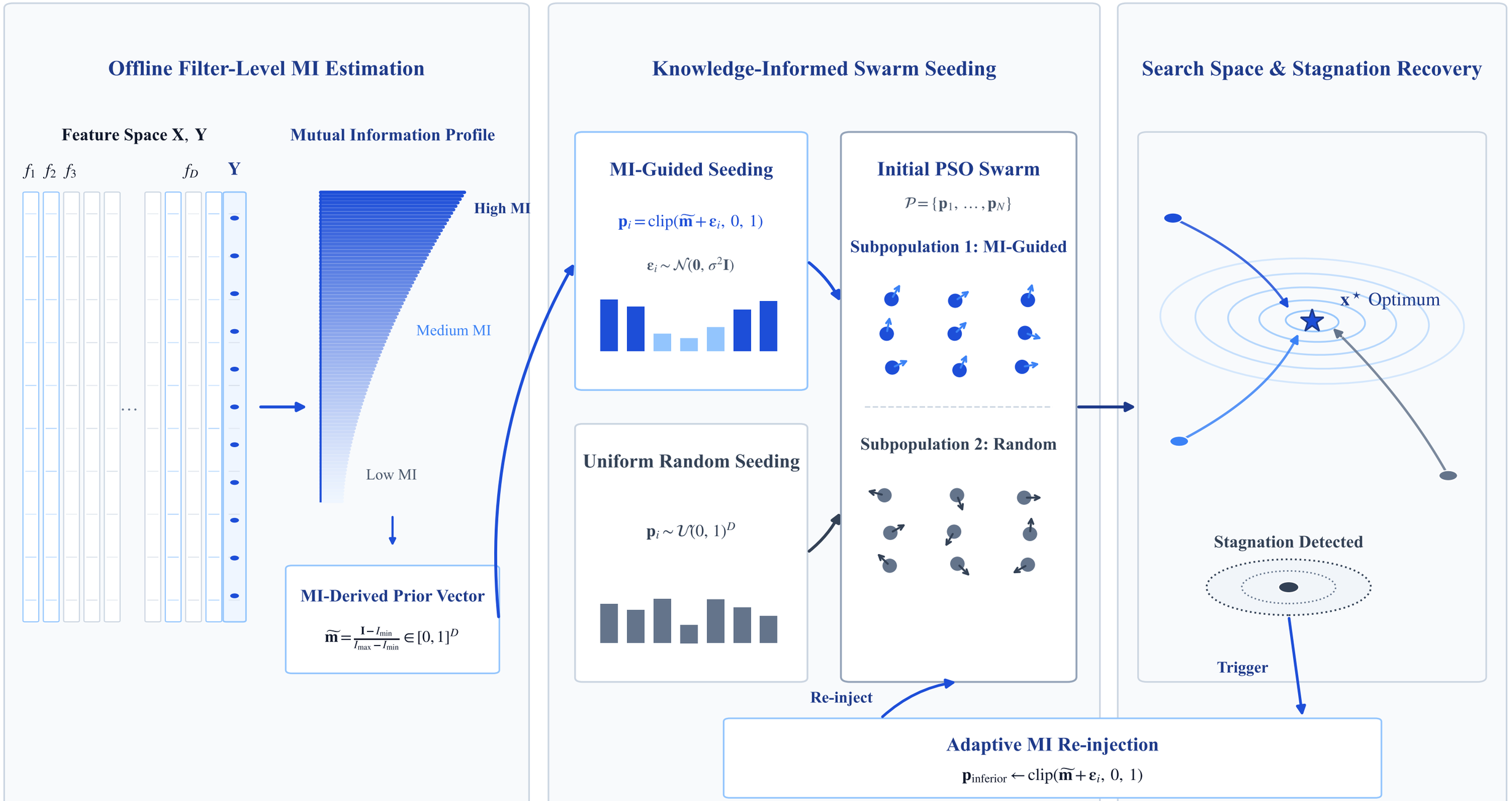

Offline Filter-Level MI Estimation
Feature Space X, Y
f_1 f_2 f_3 ... f_D Y
Mutual Information Profile
High MI
Medium MI
Low MI
MI-Derived Prior Vector
$\widetilde{\mathbf{m}} = \frac{\mathbf{I} - I_{\min}}{I_{\max} - I_{\min}} \in [0,1]^D$
Knowledge-Informed Swarm Seeding
MI-Guided Seeding
$\mathbf{p}_i = \mathrm{clip}(\widetilde{\mathbf{m}} + \boldsymbol{\varepsilon}_i,\ 0,\ 1)$
$\boldsymbol{\varepsilon}_i \sim \mathcal{N}(\mathbf{0},\ \sigma^2\mathbf{I})$
Uniform Random Seeding
$\mathbf{p}_i \sim \mathcal{U}(0,\ 1)^D$
Initial PSO Swarm
$\mathcal{P} = \{\mathbf{p}_1,\ \ldots,\ \mathbf{p}_N\}$
Subpopulation 1: MI-Guided
Subpopulation 2: Random
Re-inject
Adaptive MI Re-injection
$\mathbf{p}_{\text{inferior}} \leftarrow \mathrm{clip}(\widetilde{\mathbf{m}} + \boldsymbol{\varepsilon}_i,\ 0,\ 1)$
Search Space & Stagnation Recovery
$\mathbf{x}^\star$ Optimum
Stagnation Detected
Trigger